

Chapter 1

Single-Answer Questions

1.1 Chapter 1: Network Security Concepts and Specifications

Question 1

Topic: CIA - Confidentiality

Type: Single Choice

Which security attribute ensures that sensitive information is not disclosed to unauthorized entities or eavesdropped during transmission?

- A. Integrity
- B. Confidentiality
- C. Availability
- D. Non-repudiation

Correct Answer: B

Concept & Rationale: Confidentiality ensures that information can be accessed and viewed only by authorized personnel. Cryptographic encryption is the primary mechanism to enforce confidentiality across untrusted networks.

Question 2

Topic: CIA - Integrity

Type: Single Choice

Which security property guarantees that transmitted data has not been altered, deleted, or forged in transit?

- A. Availability
- B. Integrity
- C. Confidentiality
- D. Authenticity

Correct Answer: B

Concept & Rationale: Integrity guarantees that data has not been tampered with or modified during transit or storage. Cryptographic hash functions (such as SHA-256) and message authentication codes are primary mechanisms to verify integrity.